

Tissue Growth under Large Phenotypic Heterogeneity

A Finite-Volume Model for Competing Cell Populations

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Abstract

We simulate the growth of tissue in which several cell types move, divide and compete for space at the same time. The model follows Debiec, Mandal and Schmidtchen (2025): every cell moves along the negative gradient of a shared pressure (Darcy’s law) and grows logistically until the tissue is locally full. We derive the model, discretize it with a finite-volume scheme, and show simulations for N species in one dimension and for two species in two dimensions. In both settings, initially mixed populations separate over time into spatially distinct regions.

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1 Introduction

Simulating how tissue develops helps to predict and understand it, which matters in tumor research. We present a model that captures several cell types with different growth properties. In d -dimensional space we solve a system of partial differential equations whose solution is the density of each of the N phenotypes over time t .

We first explain the model. We then present an implementation in one dimension for N species and afterwards show results for two species in two dimensions. Both implementations solve the system with the finite-volume method.

2 The mathematical model

We explain the tissue-growth model of D ebiec, Mandal and Schmidtchen (2025). We consider N phenotypes and treat time as continuous, then discretize it in the implementation. We keep the spatial dimension $d \in \mathbb{N}$ general.

2.1 Pressure and velocity

We first study how the pressure at different points drives the motion, that is the velocity and direction. We start from a mass with the same properties everywhere except for the pressure $p: \mathbb{R}^d \rightarrow \mathbb{R}$. The motion is a vector field $v: \mathbb{R}^d \rightarrow \mathbb{R}^d$ that assigns a direction and a speed $\|v\|$ to each point. Darcy’s law links the two:

$$v(x) = -\nabla p(x). \quad (1)$$

The negative gradient of p points in the direction of steepest descent and is orthogonal to the level sets of p . The mass therefore moves from the highest pressure toward the lowest, as expected.

This model stays simplified. It says nothing about how the pressure evolves, because v and p do not depend on time t . We also assume a viscosity $\nu = 0$, so the mass moves without delay. We keep this assumption because it simplifies both the equations and the implementation.

2.2 The model for finitely many species

2.2.1 Notation

Let $N \in \mathbb{N}$ be the number of species. We write $n^{(i)}(x, t)$ for the density of species i at point $x \in \mathbb{R}^d$ and time $t \geq 0$, for each $i \in \{1, \dots, N\}$. The average density at x and t is

$$\bar{n}(x, t) := \frac{1}{N} \sum_{i=1}^N n^{(i)}(x, t). \quad (2)$$

For brevity we drop the arguments (x, t) from $n^{(i)}$ and $\bar{n}$.

2.2.2 Differential equation

For each phenotype $i \in \{1, \dots, N\}$ the density evolves according to

$$\frac{\partial n^{(i)}}{\partial t} = \nabla \cdot (n^{(i)} \nabla \bar{n}) + n^{(i)} G^{(i)}(\bar{n}). \quad (3)$$

The operator $\nabla = (\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_d})$ collects the d partial derivatives. The term $\nabla \bar{n}$ is the overall motion of all cells under Darcy's law. Multiplied by the density $n^{(i)}$ it gives the actual velocity of species i , and the divergence $\nabla \cdot (n^{(i)} \nabla \bar{n})$ states where the density of that phenotype grows or shrinks.

If no cells are born and none die, the second term vanishes and pure transport remains:

$$\partial_t n^{(i)} = \nabla \cdot (n^{(i)} \nabla \bar{n}). \quad (4)$$

2.2.3 The growth rate

The second term describes the behaviour of the cell types through the growth function $G^{(i)}$. We define $G^{(i)}$ separately for each phenotype; it depends on i and on the average density $\bar{n} \in \mathbb{R}$. Let $G^{(i)}: \mathbb{R}_+ \rightarrow \mathbb{R}$, $\bar{n} \mapsto G^{(i)}(\bar{n})$ with the following properties:

- (G1) *Continuously differentiable*: $G^{(i)} \in C^1(\mathbb{R})$.
- (G2) *Strictly decreasing*: $\exists \alpha > 0$ with $\partial_{\bar{n}} G^{(i)}(\bar{n}) \leq -\alpha$ for all $\bar{n} \in \mathbb{R}$.
- (G3) *Homeostatic pressure*: for each i there is a density $n_{(i)}^* > 0$ with $G^{(i)}(n_{(i)}^*) = 0$. We call the largest of these values $\bar{n}^* := \max_i n_{(i)}^*$. It depends on neither x nor t and is constant for each i .

The strict monotonicity (G2) reflects that cells divide less easily the higher the surrounding pressure, since less space remains for new tissue. The homeostatic pressure (G3) is the point where death and division rates balance. Since $n^* > 0$, every species grows at a sufficiently small density, and the monotonicity gives exactly one root. Negative growth turns positive again only once the pressure has dropped enough.

The growth rate takes the average density $\bar{n}$ as its argument and is multiplied by the density of the species at that point. This gives the absolute amount of growth, which is added to the spatial motion and enters the change in the density over time.

2.2.4 Initial condition

To simulate the process we need an initial distribution for each phenotype. Let $n^{(i),\text{in}}: \mathbb{R}^d \rightarrow [0, \infty)$ be the initial distribution with the following properties:

- (I1) *L^1 and L^∞ integrability*: $n^{(i),\text{in}} \in L^1(\mathbb{R}^d) \cap L^\infty(\mathbb{R}^d)$.
- (I2) *Boundedness*: $n^{(i),\text{in}}(x) \leq \bar{n}^*$ (see (G3)) for Lebesgue-almost every $x \in \mathbb{R}^d$.
- (I3) *Centrality*: $\int_{\mathbb{R}^d} n^{(i),\text{in}}(x) \|x\|^2 dx < \infty$.

The integral of a density is the total number of cells of that type, so L^1 requires finitely many cells. L^∞ bounds the density on every subset of positive Lebesgue measure, which is also physically reasonable. (I2) tightens this with the upper bound $\bar{n}^*$ from (G3): at least one phenotype has a non-negative growth rate everywhere, so the system does not start overcrowded. (I3) weights the density by $\|x\|^2$, so points near the origin count little and distant points count strongly. A finite integral keeps most cells near the origin at the start; they spread out later under Darcy's law.

Together with given $G^{(i)}$ and $n^{(i),\text{in}}$, we obtain for all $i \in \{1, \dots, N\}$ the final system

$$\begin{aligned} \partial_t n^{(i)} &= \nabla \cdot (n^{(i)} \nabla \bar{n}) + n^{(i)} G^{(i)}(\bar{n}), \\ n^{(i)}(x, 0) &= n^{(i),\text{in}}(x). \end{aligned} \quad (5)$$

3 Numerical implementation of the 1D N -species model

We implement the Darcy model of Section 2 for finitely many species in one spatial dimension. The focus is on the discretization, the model assumptions and the interpretation of the results.

3.1 A discretization example

As an illustration we take $d = 1$ with $N = 2$ species on $\Omega = [0, 3]$, split coarsely into three control volumes of width $\Delta x = 1$:

$$C_1 = [0, 1], \quad C_2 = [1, 2], \quad C_3 = [2, 3].$$

As initial densities we choose the piecewise-constant cell averages

$$n^{(1)} = (0.8, 0.2, 0), \quad n^{(2)} = (0, 0.4, 0.6),$$

which give the average density $\bar{n}_j = \frac{1}{2}(n_j^{(1)} + n_j^{(2)}) = (0.4, 0.3, 0.3)$. In 1D the velocity field follows from $v = -\partial_x \bar{n}$. Discrete differences at the cell faces give

$$v_{1 \rightarrow 2} \approx -(\bar{n}_2 - \bar{n}_1) = 0.1, \quad v_{2 \rightarrow 3} \approx 0.$$

Mass therefore flows from C_1 to C_2 , while no transport happens between C_2 and C_3 . This shows the pressure balancing from high to low average density.

For the growth we take $G^{(1)}(\bar{n}) = 1 - \bar{n}$ and $G^{(2)}(\bar{n}) = 0.5 - \bar{n}$. The local reaction term $n_j^{(i)} G^{(i)}(\bar{n}_j)$ lets both species grow where they are present, since $\bar{n}_j < 1$ and $\bar{n}_j < 0.5$ in every cell. If $n_j^{(i)} = 0$, no new species appears. The full dynamics

$$\partial_t n^{(i)} = \partial_x (n^{(i)} \partial_x \bar{n}) + n^{(i)} G^{(i)}(\bar{n})$$

move existing mass along the gradient of the total density and amplify or damp the local density depending on the overall occupancy. This minimal example shows the two central mechanisms: pressure-driven transport from high to low average density, and growth that $\bar{n}$ regulates locally. In the next step we treat both effects together per time step.

3.2 Discretization and parameters

The domain is the interval $[0, L]$, divided into J equidistant control volumes of width $\Delta x = L/J$. The unknowns are the cell densities $n_j^i(t) \approx n_i(x_j, t)$ of the N species at the cell centers x_j . We simulate up to a final time T with a fixed step size Δt . We choose the pressure as

$$p(\rho) = \rho^\gamma, \quad \rho = \sum_{i=1}^N n_i, \tag{6}$$

where $\gamma > 1$ produces a steeply rising pressure at high total densities. The species differ only in their constant growth rates r_i . For this we discretize an interval $[r_{\min}, r_{\max}]$ uniformly, permute it at random and assign the values to the species. This creates heterogeneous but time-constant growth conditions. As initial data we consider two variants, a strictly separated distribution and overlapping Gaussian profiles with maxima spread along the interval. In both cases we scale so that $\rho(x, 0) \leq 1$ for all x , so the maximum packing density already holds at the start.

3.3 Finite-volume scheme

Each species satisfies

$$\partial_t n_i + \partial_x(n_i v) = n_i r_i(1 - \rho), \quad v = -\partial_x p(\rho). \quad (7)$$

We discretize this equation with an explicit finite-volume scheme. It rests on the balance over the control volumes and therefore conserves total mass at the discrete level; Carrillo, Chertock and Huang (2015) give a detailed account. From the densities we first compute the total density $\rho_j = \sum_i n_j^i$ and the pressure $p_j = (\rho_j)^\gamma$. The velocity at the cell faces then follows from a discrete derivative,

$$v_{j+\frac{1}{2}} = -\frac{p_{j+1} - p_j}{\Delta x}, \quad (8)$$

with no-flux boundary conditions $v_{1/2} = v_{J+1/2} = 0$. Once the velocity is known, we discretize the transport with an upwind flux that takes the density from the upwind cell:

$$F_{i,j+\frac{1}{2}} = \begin{cases} v_{j+\frac{1}{2}} n_j^i, & v_{j+\frac{1}{2}} \geq 0, \\ v_{j+\frac{1}{2}} n_{j+1}^i, & v_{j+\frac{1}{2}} < 0. \end{cases} \quad (9)$$

We set the reaction term locally as $R_j^i = n_j^i r_i(1 - \rho_j)$; it gives logistic growth with carrying capacity $\rho = 1$. An explicit time step thus has four parts: pressure and velocity, fluxes, reaction term, and the density update. We clip negative values numerically. A sufficiently small Δt ensures stability.

3.4 Model assumptions

The 1D simulation rests on four assumptions: the tissue is one-dimensional; all species experience the same pressure field, which depends on the total density; the local density is bounded by $\rho \leq 1$; and we use insulating boundary conditions. Section 5 gives a full overview of all assumptions of the model and both implementations.

3.5 Visualization and phase separation

For selected times we show two views: the absolute density of one species together with the total density, and the relative fraction of one species in the total population.

Case 1: strictly separated initial distribution. Two species start in disjoint regions, so competition first occurs only at the interface. The species with the higher growth rate builds up more local pressure and displaces the slower one. After the total density reaches its maximum, a stable, spatially separated configuration sets in (Figure 1).

Case 2: overlapping Gaussian profiles. Several species start with overlapping Gaussian profiles and therefore in direct competition. In the same spatial regions, faster-growing species reach higher densities and displace slower ones. Over time a clear phase separation into distinct domains emerges, while the total density reaches the bound $\rho = 1$ (Figure 2).

4 Numerical implementation of the 2D two-species model

4.1 Structure and parameters

We implement the simulation in Python as the class `TissueSim2D`. All attributes can be changed and all methods called flexibly, which eases parameter studies and extensions. The class provides

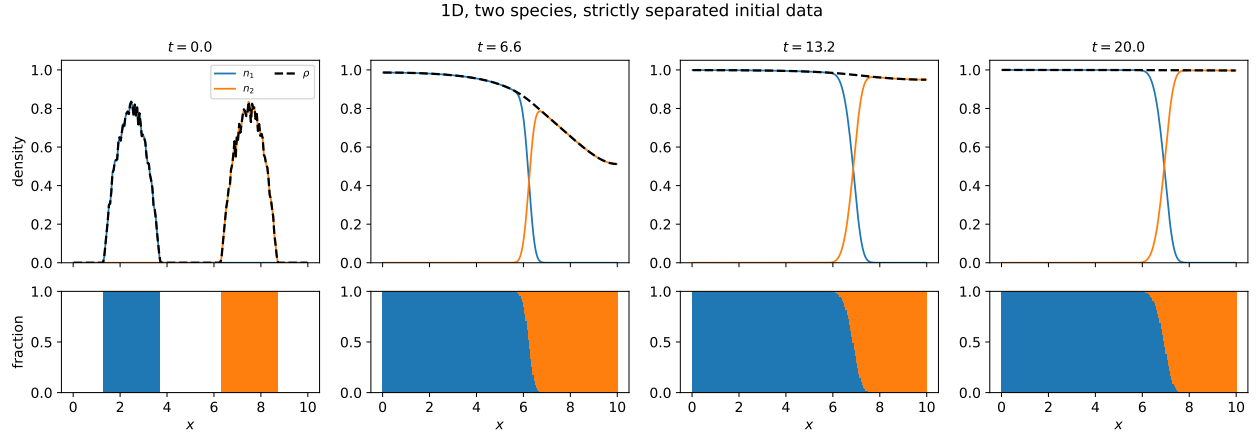


Figure 1: Case 1: evolution of the cell distribution for a strictly separated initial distribution.

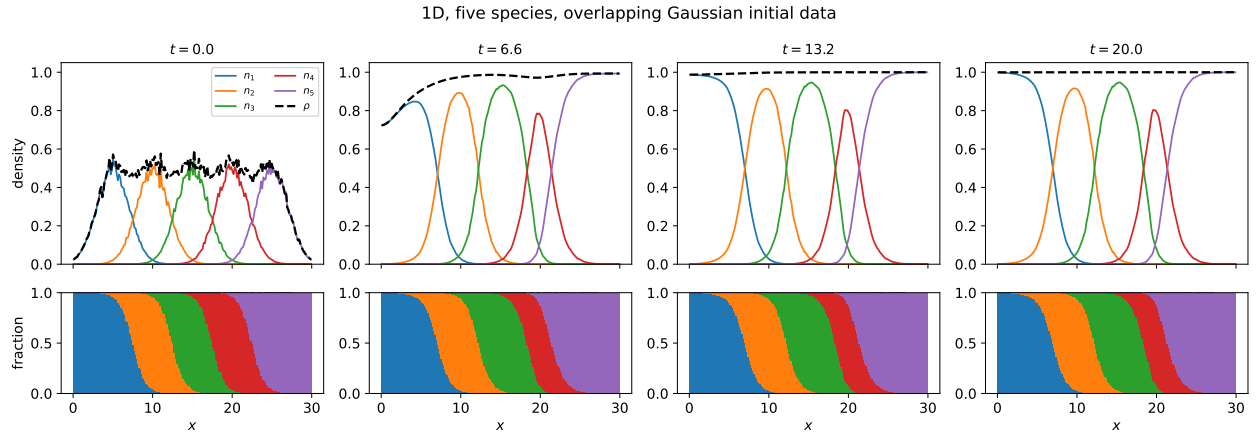


Figure 2: Case 2: evolution of the cell distribution for overlapping Gaussian bells.

the following methods:

- `__init__`: initializes the grid, the model parameters and the initial densities.
- `update_n`: computes the new densities from those of the previous step with a finite-volume scheme and explicit Euler.
- `run`: runs the simulation over all steps and stores snapshots at regular intervals.
- `visualize`: shows both species, the total density and a direct comparison.
- `save / load`: store or load the full simulation state.

The main parameters are the grid size `grid_size` (the grid has `grid_size` $\times$ `grid_size` cell-centered control volumes), the number of steps `count_timesteps`, the step size `dt`, the snapshot interval `snapshot_every` and the boundary type `boundary` (`neumann` or `dirichlet`). The species behaviour is set by the growth rates `G1`, `G2`, the competition coefficients `comp1`, `comp2` (growth inhibition by the other species), the pressure response `beta1`, `beta2` and the self-diffusion `eps1`, `eps2`. The initial distribution is fixed by `center`, `peak`, `sigma` and `background` for both species. The default parameters are symmetric, so the influence of a single parameter can be studied in isolation.

4.2 Finite-volume scheme and time discretization

4.2.1 Equations

The model consists of two coupled conservation laws for $i, j \in \{1, 2\}$ with $i \neq j$:

$$\partial_t n_i + \nabla \cdot (n_i v_i) = \varepsilon_i \Delta n_i + G_i n_i (1 - n_i - c_i n_j), \quad v_i = -\beta_i \nabla \bar{n}, \quad \bar{n} = \frac{1}{2}(n_1 + n_2). \quad (10)$$

Compared with the 1D model, two effects are added: the self-diffusion $\varepsilon_i \Delta n_i$ and the asymmetric competition through the coefficients c_i .

4.2.2 Spatial discretization

The densities $n_i^{k,l}$ are cell-centered averages on a uniform grid with $0 \leq k, l \leq N-1$; we evaluate the fluxes at the faces $k \pm \frac{1}{2}, l$ and $k, l \pm \frac{1}{2}$. We first approximate the face velocities with finite differences,

$$v_{i,x}^{k+\frac{1}{2},l} = -\beta_i \frac{n^{k+1,l} - n^{k,l}}{\Delta x}, \quad v_{i,y}^{k,l+\frac{1}{2}} = -\beta_i \frac{n^{k,l+1} - n^{k,l}}{\Delta y}. \quad (11)$$

From these we form the advective flux $F = vn$ with the density taken upwind,

$$F_x^{k+\frac{1}{2},l} = \begin{cases} v_x^{k+\frac{1}{2},l} n^{k,l}, & v_x \geq 0, \\ v_x^{k+\frac{1}{2},l} n^{k+1,l}, & v_x < 0, \end{cases} \quad F_y^{k,l+\frac{1}{2}} = \begin{cases} v_y^{k,l+\frac{1}{2}} n^{k,l}, & v_y \geq 0, \\ v_y^{k,l+\frac{1}{2}} n^{k,l+1}, & v_y < 0. \end{cases} \quad (12)$$

The divergence follows from the difference of the face fluxes,

$$(\nabla \cdot F)^{k,l} = \frac{F_x^{k+\frac{1}{2},l} - F_x^{k-\frac{1}{2},l}}{\Delta x} + \frac{F_y^{k,l+\frac{1}{2}} - F_y^{k,l-\frac{1}{2}}}{\Delta y}. \quad (13)$$

The local growth term is $R_i^{k,l} = G_i n_i^{k,l} (1 - n_i^{k,l} - c_i n_j^{k,l})$.

4.2.3 Time discretization and stability

Once the flux divergence and the reaction term are known, we integrate with an explicit Euler scheme:

$$n_i^{k,l}(t + \Delta t) = n_i^{k,l}(t) + \Delta t \left(-(\nabla \cdot F_i)^{k,l} + \varepsilon_i(\Delta n_i)^{k,l} + R_i^{k,l} \right). \quad (14)$$

The explicit scheme is subject to stability limits, a CFL condition for the advection and a diffusion condition of the form $\Delta t \lesssim C\Delta x^2/\varepsilon_i$. We choose Δt empirically so that the solutions stay stable in the parameter ranges considered; we do not carry out a formal stability analysis.

4.2.4 Boundary conditions

We treat the boundary in two ways. The Neumann condition extends the boundary values by first-order constant extrapolation from the interior, so the normal flux vanishes and total mass is conserved. The Dirichlet condition sets all boundary cells to zero after each step and models a hostile environment in which cells at the boundary are removed immediately.

4.3 Visualization of the results

The visualization shows four heatmaps: the density of the first species, that of the second, the total density and a direct comparison. A slider sets the time, and a checkbox toggles the contours. Red stands for species 1, blue for species 2.

Default run and variations. Starting from the same initial state, we change individual parameters. A higher growth rate (`G1` = 1.3) lets species 1 spread faster and displace species 2. A stronger pressure response (`beta1` = 3.0) moves species 1 noticeably faster than species 2. Changing several parameters at once lets us compare very different cell types and observe their behaviour in the model.

5 Overview of all assumptions

Table 1 summarizes the assumptions on which the model and its two implementations rest.

References

- [1] J. A. Carrillo, A. Chertock, and Y. Huang. A finite-volume method for nonlinear nonlocal equations with a gradient flow structure. *Communications in Computational Physics*, 17(1):233–258, 2015.
- [2] T. Dębiec, M. Mandal, and M. Schmidtchen. From finite to continuous phenotypes in (visco-)elastic tissue growth models. *Journal of Differential Equations*, 437:113375, 2025.

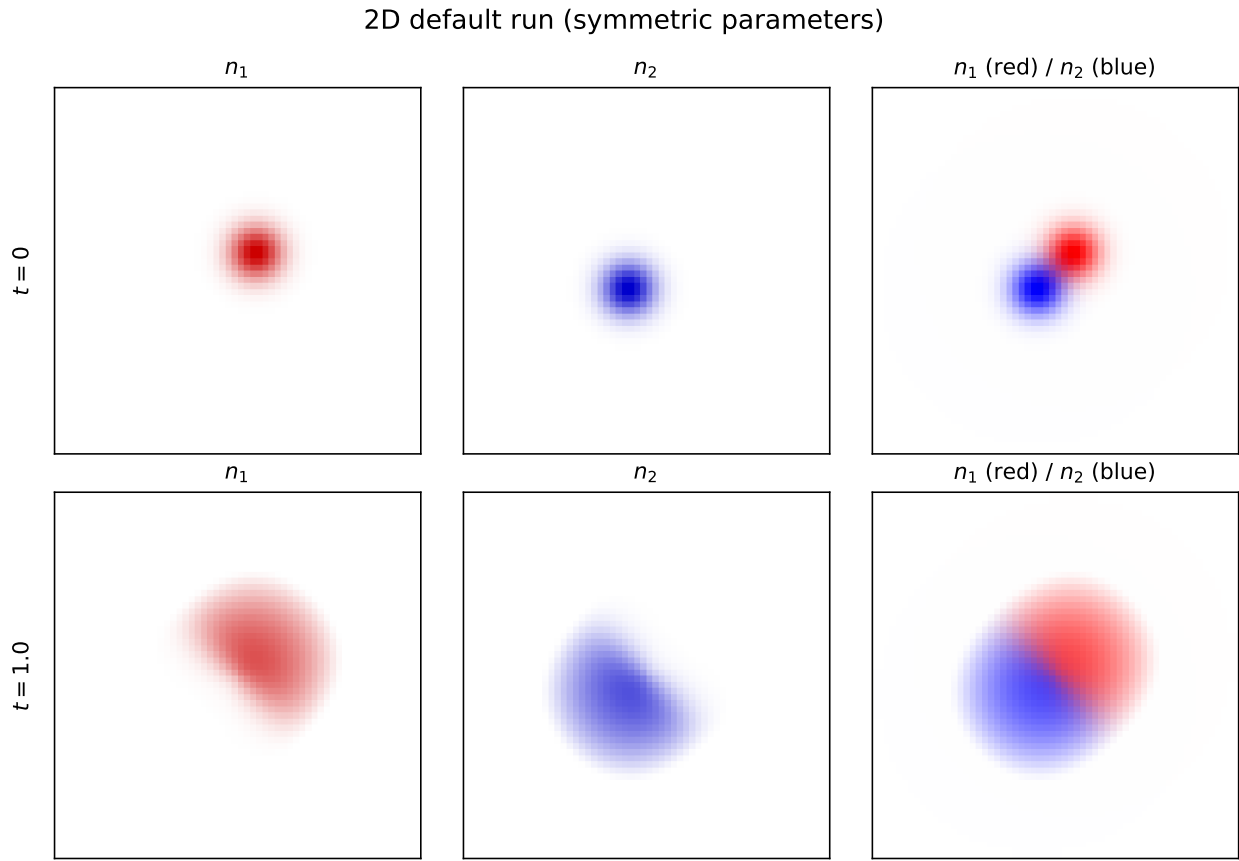


Figure 3: Default run with symmetric parameters: initial and developed state. Red is species 1, blue is species 2.

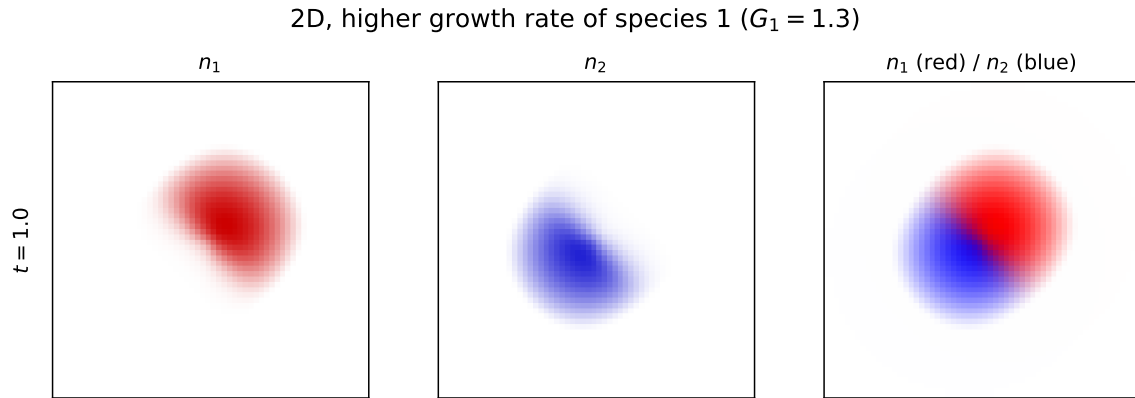


Figure 4: Higher growth rate of species 1 ($G_1 = 1.3$): species 1 grows faster and displaces species 2 more.

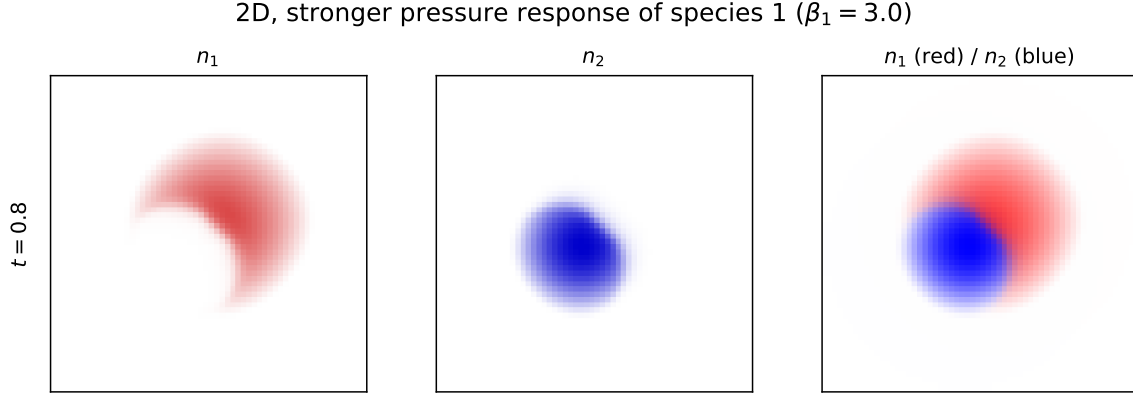


Figure 5: Stronger pressure response of species 1 ($\beta_1 = 3.0$): species 1 spreads out much faster than species 2.

Area	Assumption
Motion (Darcy)	Velocity $v = -\nabla p$; viscosity $\nu = 0$, so motion is instantaneous.
Pressure	Shared pressure field $p(\rho) = \rho^\gamma$ with $\gamma > 1$, depending only on the total density ρ .
Growth (G1–G3)	$G^{(i)} \in C^1$, strictly decreasing ($\partial_{\bar{n}} G^{(i)} \leq -\alpha$), with a single homeostatic pressure $n_{(i)}^* > 0$.
Initial data (I1–I3)	$n^{(i),\text{in}} \in L^1 \cap L^\infty$, bounded by $\bar{n}^*$, with a finite second moment (central initial distribution).
Saturation	Maximum local density $\rho \leq 1$, already satisfied at the start.
Geometry	1D model on $[0, L]$ or 2D model on the unit square.
Boundary	Insulating (Neumann) conditions; Dirichlet as an alternative in 2D.
Time	Explicit Euler; Δt chosen empirically below the CFL and diffusion limits.
2D addition	Self-diffusion $\varepsilon_i \Delta n_i$ and asymmetric competition c_i between the two species.

Table 1: Overview of the model and implementation assumptions.